BLG 561E

01.12.2020

Deep Learning

Week 6-7

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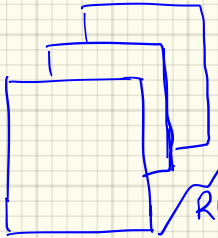
CONVOLUTIONAL NEURAL NETWORKS (CNNs)

→ Only the hypothesis functions change → ① component.
② & ③ ✓

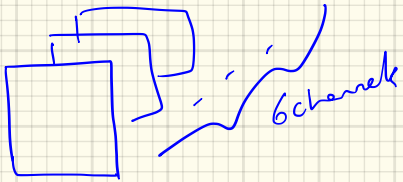
Say you have large (spatial) data : FCN



F.CN.

RGB → 3 channels

$32 \times 32 \times 3 \Rightarrow 3072 \times 1$

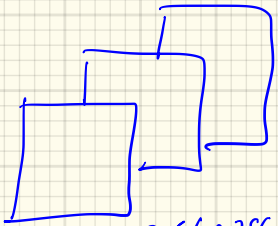


6 channels

$28 \times 28 \times 6 = 4704$

$W^1 : 3072 \times 4704 \approx$ 14 Million!

14M parameters



$\Rightarrow$ FCNs would need a very large # of parameters.

$256 \times 256 \times 3 = 3 \cdot 2^{16} \approx 600K$ dim.

→ CNNs : can handle large spatial data ; to do that restrict the weights in 2 ways :

1) Require that activations between layers occur in a "local" manner (sparsity)

2) Require that all activations share the same weight (parameter sharing)

New hypothesis for CNNs (1st component changed only) :

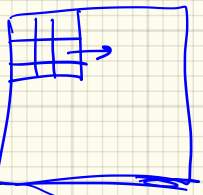
(1) $z^{i+1} = w^i * z^i + b^i$

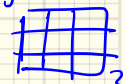
$a^{i+1} = f(w^i * a^i + b^i)$

→ maxpool, BN : f: RELU or other nonlinearity

*: convolution operation

e.g. w^i





3x3

Compare to FCN :

$z^{i+1} = \underline{w^i} \cdot \underline{z^i} + b^i$

$a^{i+1} = f(z^{i+1})$

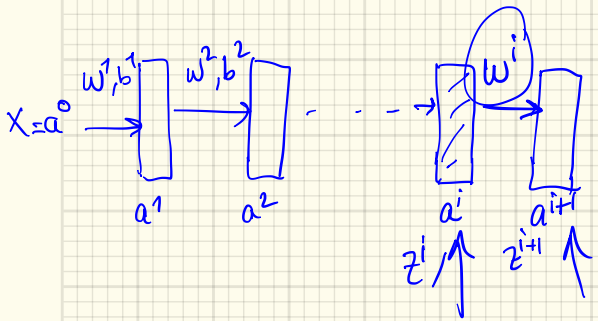
Recaps : CNNs : New hypothesis :

motivations ① sparsity, locality
② shared connections

$$\boxed{\begin{aligned} z^{i+1} &= W^i * a^i + b^i \\ a^{i+1} &= f(z^{i+1}) = f(W^i * a^i + b^i) \end{aligned}}$$

$*$: convolution operator

; f : e.g. RELU nonlinearity



— maxpool
— BN ...
} (optional) other components

Recall FC layer

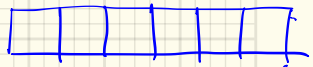
$$a^{i+1} = f(W^i \cdot a^i + b^i)$$

Write convolution as a matrix operator:

$$\underline{\underline{W^i}} = \begin{bmatrix} w_1 & w_2 & w_3 & 0 & 0 & 0 \\ 0 & w_1 & w_2 & w_3 & 0 & 0 \\ 0 & 0 & w_1 & w_2 & w_3 & 0 \\ 0 & 0 & 0 & w_1 & w_2 & w_3 \end{bmatrix}_{4\times 6}$$

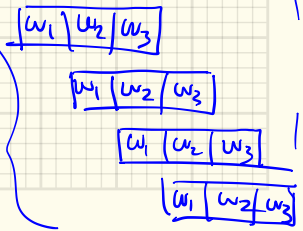
say $W^i = (w_1, w_2, w_3)$ 1D filter for illustration

↳ Input $a^i_{6\times 1}$



$W^i_{3\times 1}$

↳ 4 possible "valid" shifts



$$\Rightarrow \boxed{a^{i+1} = f(\underline{\underline{W}}^i \cdot a^i + b^i)}$$

What happens at backprop? Recall backprop for FC

upstream gradient:

$$\frac{\partial l}{\partial a^i} = g^i = (W^{i+1})^T g^{i+1} \cdot f'(z^{i+1})$$

let's transpose the convolution matrix:

$$\Rightarrow \underline{\underline{W}}_i^T = \begin{bmatrix} w_1 & 0 & 0 & 0 \\ w_2 & w_1 & 0 & 0 \\ w_3 & w_2 & w_1 & 0 \\ 0 & w_3 & w_2 & w_1 \\ 0 & 0 & w_3 & w_2 \\ 0 & 0 & 0 & w_3 \end{bmatrix}_{6\times 4}$$

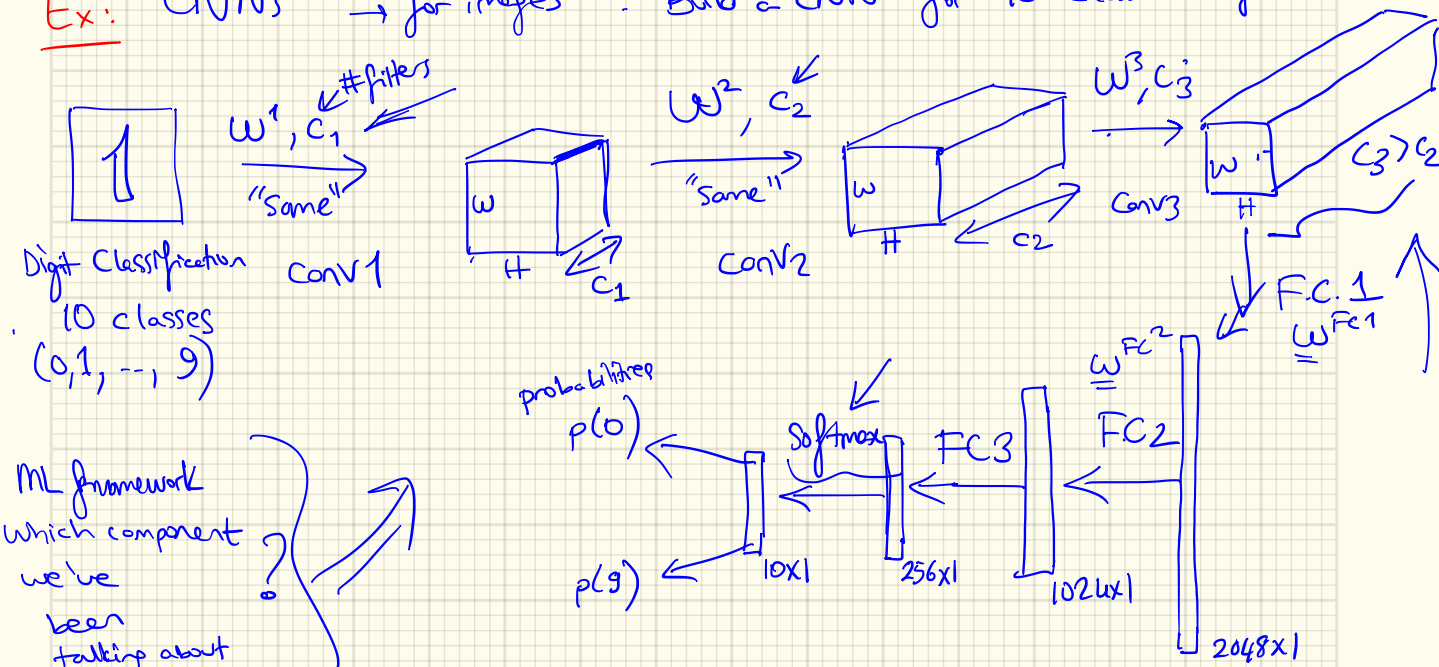
$g^{i+1}: 4\times 1$

filter is flipped.

So in CNNs, backprop just flips convolutions

2D Conv: flip vertically & horizontally.

Ex: CNNs → for images : Build a CNN for 10-class classification

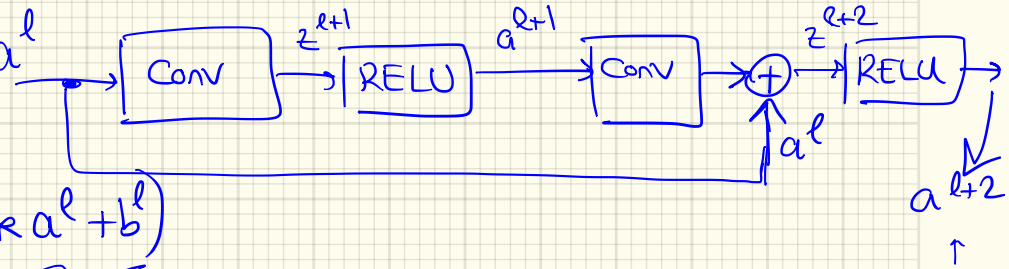


ML framework which component we've been talking about ?

1. this whole model defines our hypothesis function
2. loss →
3. pick an optimization →

Ex: ResNet:

1 Residual Block:



$$a^{l+1} = f(\underbrace{w^l * a^l + b^l}_{z^{l+1}})$$

$$a^{l+2} = f(z^{l+2} + a^l) = f\left(f(w^{l+1} * a^{l+1} + b^{l+1}) + a^l\right)$$